NNN-T-001293 (DSA-DM)
November 4, 1969

INTERIM FEDERAL SPECIFICATION

TUBE, BLOOD COLLECTING, Vacuum, Nonsterile

This Interim Federal Specification was developed by the Defense Personnel Support Center, 2800 S. 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers nonsterile, vacuum, blood collecting tubes, suitable for collecting, containing and transporting blood samples.

1.2 Classification. Nonsterile, vacuum, blood collecting tubes shall be of the following types, sizes, color codes and grades, as specified (see 6.1).

Type I - With anticoagulant.
Type II - Without anticoagulant.

Size 1 - 5 milliliters (ml.) nominal aspiration.
Size 2 - 7 milliliters (ml.) nominal aspiration.
Size 3 - 15 milliliters (ml.) nominal aspiration.

Color codes

Red
Blue
Gray
Lavender
Black

Grade A - Ammonium Oxalate and Potassium Oxalate as additive.
Grade B - Lithium Oxalate as additive.
Grade C - Thymol and Sodium Fluoride as additive.
Grade D - Sodium Fluride as additive.
Grade E - Disodium Ethylenediaminetetracetate Dihydrate as additive.
Grade F - Sodium Oxalate as additive.

FSC 6630

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2. APPLICABLE DOCUMENTS

2.1 Specifications and Standards. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

DD-G-541	Glass, Laboratory.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

STANDARDS

Federal

Test Method Std. No. 601	Rubber, Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements, as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this Specification and other product specifications required by activities outside Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston; New York; Atlanta; Chicago; Kansas City, Mo.; Fort Worth; Denver; San Francisco; Auburn, Washington; and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

SPECIFICATIONS

Military

MIL-L-10547	Liners, Case and Sheet Overwrap; Water-Vapor-proof or Waterproof, Flexible.

STANDARDS

Military

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MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 Marking for Shipment and Storage.

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN PHARMACEUTICAL ASSOCIATION PUBLICATION

National Formulary (N.F.)

U.S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States (U.S.P.)

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pennsylvania 18042.)

REAGENT CHEMICALS AND STANDARDS WITH METHODS OF TESTING AND ASSAYING
By Joseph Rosin

(Application for copies should be addressed to D. Van Nostrand Company, Inc., Princeton, New Jersey 08540.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Glass. Glass shall be in accordance with Federal Specification DD-G-541, type II.

3.1.2 Rubber. The rubber formulation shall contain at least seventy percent (70%) by volume of isobutylene-isoprene polymer and at least five percent (5%) by weight of zinc oxide. The rubber shall be nonblooming, nonbleeding and self sealing.

3.2 Construction. The blood collecting tubes shall consist of a tube and a stopper. The stoppers shall fit snugly in the tubes to form an air-tight seal.

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3.2.1 Tube. The tube shall be transparent, shall be fabricated of glass and shall be in accordance with Figure 2. Wall thickness shall be uniform. Bottom shall be round and open end shall be straight without a bead or a flare.

3.2.2 Stopper. Stopper shall be fabricated of rubber and shall be in accordance with Figure 1 and applicable portions of Figure 2. Stopper shall be a diaphragm, molded in one piece. Top of diaphragm shall be flat with a recessed center and bottom shall be a hollow plug. Coloring material shall be inorganic and nonbleeding. Shore durometer reading of stopper shall be 50 ± 5 when tested as specified in 4.4.1.

3.3 Color coding. Additives in tubes shall be differentiated by means of colored stoppers. Color of stopper and additive shall correspond as specified in Table I.

Table I. Stopper Color and Additives

Stopper Color	Grade	Additive (Note at end of table shall apply)
Red	None	None
Blue	A	6.0 milligrams U.S.P. Ammonium Oxalate $(NH_4)_2\ C_2O_4 \cdot H_2O$ and 4.0 mg. N.F. Potassuim Oxalate $K_2C_2O_4 \cdot H_2O$
Gray	B	22.5 mg, 15.0 mg minimum N.F. Lithium Oxalate LiC_2O_4.
	OR	OR
	C	5 mg. N.F. Thymol $C_{10}Hi_4O$ and 50 mg. USP Sodium Fluoride, NaF
	OR	OR
	D	27.0 to 34.5 mg USP Sodium Fluoride NaF
Lavender	E	8 to 9.5 mg. Disodium Ethylenediaminetetracetate Dihydrate
Black	F	0.50 to 0.52 ml. of 0.1 Malor Solution of American Chemical Society Reagent Grade Sodium Oxalate $Na_2C_2O_4$. Water shall be distilled.

Note. Unless otherwise specified tolerances shall be plus 33-1/3 percent, minus 10 percent. Additives shall be powders except for Sodium Oxalate which shall be powder in aqueous solution. Powders shall be as specified in the applicable portions of the reagent section of the Pharmacopeia of the United States (USP) or National Formulary (NF).

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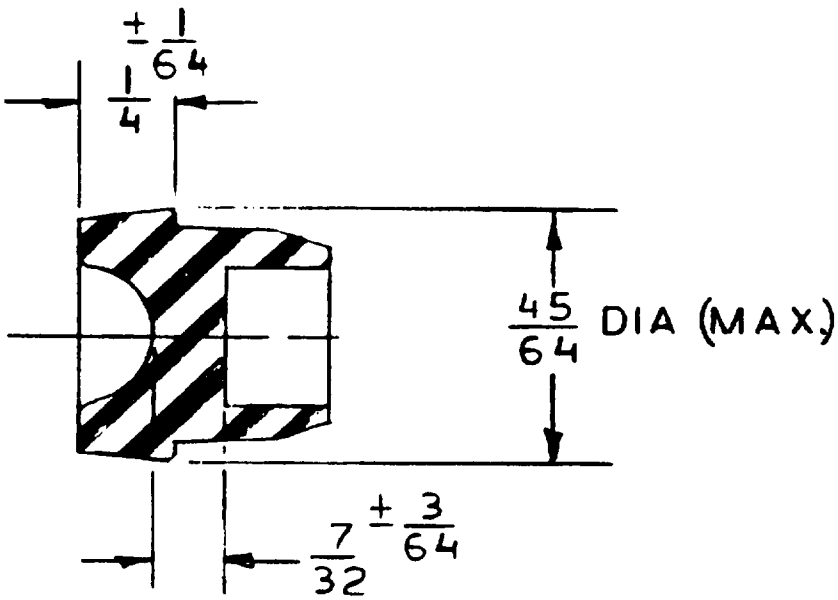
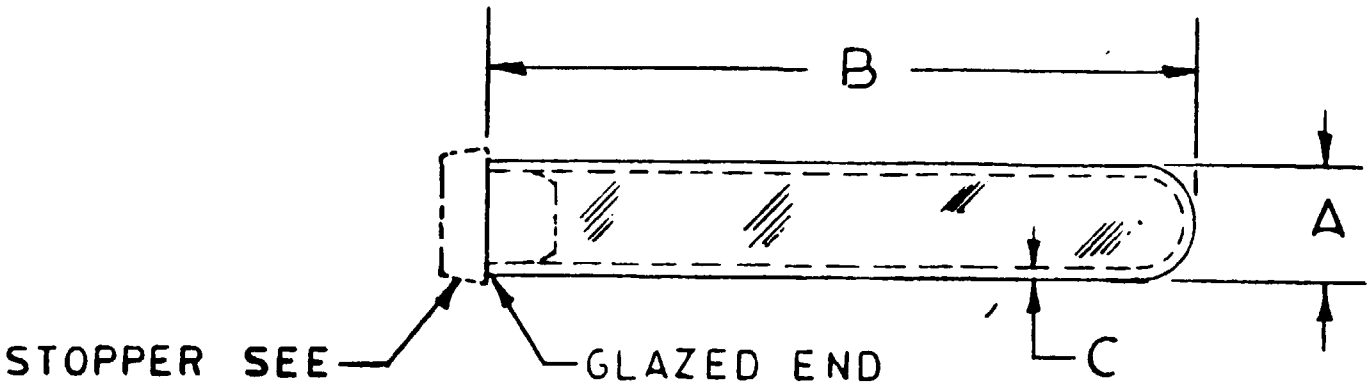


FIGURE 1 – STOPPER

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SIZE	A	B	C
1 (COLOR CODE BLACK)	12-13mm	73-77mm	0.8-1.1mm
1&2 (EXCEPT COLOR CODE BLACK)	12-13mm	98-102mm	0.8-1.1mm
3	15-16mm	125-129mm	0.9-1.1mm

FIGURE 2-TUBE AND STOPPER

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3.4 Performance

3.4.1 Vacuum. When tested as specified in 4.4.2, minimum volumes of water drawn into tubes shall be as follows:

Size 1	4.5 mls.
Size 2	6.5 mls.
Size 3	14.5 mls.

3.4.2 Resistance of tubes to centrifugal force. Resistance of tubes to centrifugal force shall be such that no breakage shall result when the tubes are tested as specified in 4.4.3.

3.5 Date of shipment. Date of shipment shall not be more than 60 days after the date of manufacture.

3.6 Expiration date. The expiration date shall be not less than 24 months from the date of manufacture. At the expiration date the blood collecting tube shall draw at least 90 percent of the volume specified in 3.4.1.

3.7 Workmanship. Workmanship shall be first class throughout. Tubes shall be free of defects listed in Table IV and other defects which detract from their appearance or may inpair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the glass and rubber, which will contain evidence of the identification of the materials required in the tube may be furnished in lieu of actual tests.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table II. Unit of product for examination purposes shall be the tube.

Table II. Sampling for Examination

Type defect	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	0.65
Minor defects	II	2.5
For dimensional examination	S-1	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III.

Table III. Sampling for Tests

Characteristic	Unit of Product	Requirement	Inspection level	AQL (percent defective)
Durometer hardness	Stopper	3.2.2	4.4.1	*
Vacuum	Tube	3.4.1	4.4.2	1.5
Resistance to centrifugal force	Tube	3.4.2	4.4.3	1.5

* Acceptance number shall be zero.

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4.3 Examination. The tubes shall be examined for defects including, but not limited to, those listed in table IV.

Table IV. Classification of Defects

Categories	Defects
Major	
101	Tube not free of cracks and chips.
102	Stopper not secured in tube.
103	Tube not free of dirt or foreign matter.
104	Tube does not contain additive (except type II)
Minor	
201	Tube not free of sharp edges.
202	Tube not free of bubbles.
203	Stopper not free of bloom.

4.3.1 Dimensional examination. The tubes shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Durometer hardness. Stoppers shall be tested in accordance with Method 3021 of Federal Test Method Standard No. 601.

4.4.2 Vacuum. A seamless steel tube Brown and Sharpe gage No. 20, 2½ ± 1/16 inches long with a point at each end and a wall thickness from 0.0055 to 0.0060 inch or 6515-864-6788, needle, shall be used for testing. One end of the testing tube or the venipuncture end of the needle shall be placed in distilled water and the other end of the testing tube or needle shall pierce the diaphragm.

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4.4.3 Resistance to centrifugal force. Tubes shall be filled to capacity with distilled water and suspended in a centrifuge which has an angle head of 50° from the vertical and a radius of 1-3/4 inches from the center of rotation to the center of the mouth of the tube shield. The centrifuge shall be accelerated gradually to a speed of 3000 revolutions per minute (r.p.m.). Centrifuge shall run at 3000 r.p.m. for 30 minutes. The current shall then be shut off and the centrifuge allowed to stop without the use of a brake.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One package containing 100 tubes, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V. Packaging Quantities

Type & Size	Unit package	Intermediate package	Exterior container
All	1 unit	NOT REQUIRED	10 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be corrugated fiberboard box of appropriate size and design fabricated of 200 pound test fiberboard. Closure of box shall be adequate to prevent spilling of contents under normal handling.

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5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with paragraph 30.4 of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621,class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Lot control number shall be shown and designated as such. Marking shall include contract or purchase order number, name of contractor, and the following legend: "STORE IN A COOL PLACE (50° F - 80° F.)" In addition, Type II Shelf-Life markings as specified in paragraph 5.2.2.1 (b) of MIL-STD-129 shall be shown. The "INSPECTION/TEST DATE" and "TEST NO." marking shall include a blank space for inclusion of data at a later date.

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5.3.2 Exterior container. Each exterior container shall be marked in accordance with MIL-STD-129. Lot control number shall be shown and designated as such. Marking shall include the legend: "STORE IN A COOL PLACE (50° F - 80° F.)." In addition, Type II Shelf-Life markings as specified in paragraph 5.2.2.1 (b) of MIL-STD-129 shall be shown. The "INSPECTION/TEST DATE" and "TEST NO." marking shall include a blank space for inclusion of data at a later date.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, name, and date of this specification.
b. Type, size, color code and grade (if applicable) required.
c. Selection of applicable packaging and packing (see 5.1 and 5.2).
d. Federal Stock Number.

6.2 This specification covers the following items appearing in the Federal Supply Catalog:

Federal Stock No.	Item Identification	Type	Size	Color Code	Grade
6630-145-1130	TUBE, BLOOD COLLECTING, Vacuum, 5 ml, with Sodium Oxalate, 100s	I	1	Black	F
6630-145-1132	TUBE, BLOOD COLLECTING, Vacuum, 5 ml, with Ammonium and Potassium Oxalate, 100s	I	1	Blue	A
6630-145-1137	TUBE, BLOOD COLLECTING, Vacuum, 7 ml without Anticoagulant, 100s	II	2	Red	None
6630-145-1143	TUBE, BLOOD COLLECTING, Vacuum, 15 ml, without Anticoagulant, 100s	II	3	Red	None
6630-145-1527	TUBE, BLOOD COLLECTING, Vacuum, 15 ml, with Lithium Oxalate, 100s	I	3	Gray	B

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Federal Stock No.	Item Identification	Type	Size	Color Code	Grade
6630-145-1533	TUBE, BLOOD COLLECTING, Vacuum, 7 ml, with Disodium Edetate, 100s	I	2	Lavender	E
6630-145-1534	TUBE, BLOOD COLLECTING, Vacuum, 7 ml, with Sodium Fluoride, 100s	I	2	Gray	D
6630-145-1535	TUBE, BLOOD COLLECTING, Vacuum, 7 ml, with Sodium Fluoride and Thymol, 100s	I	2	Gray	C

6.3 The items listed in 6.2 are to be used with the following items appearing in the Federal Supply Catalog:

Federal Stock No.	Item Identification
6515-404-2240	Needle, Hypodermic, Blood Collecting, Double Pointed, 20 Gage, 1 inch, 12s
6515-404-2260	Needle, Hypodermic, Blood Collecting, Double Pointed, 20 Gage, 1½ inch, 12s
6515-864-6788	Needle, Hypodermic, Blood Collecting, Double Pointed, Disposable, 20 Gage, 1½ inch, 100s
6630-404-2220	Holder, Blood Collecting Tube, 12s

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6.4 Specified rubber is the only known material which is compatible with sampling and analysis and provides sufficient seal to hold the vacuum over an extended period of time. Other rubber will be considered if detailed reports on the effects on sampling and analysis and the durability of the seal are submitted.

6.5 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Federal Government requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency-DM

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